Shibo Chen

Staff Architect | Chiplet NoCs, coherent fabrics, memory systems, performance modeling, and AI-assisted hardware design

Santa Clara, CA

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Summary

Staff Architect at Tenstorrent, working across chiplet-based AI/ML accelerators and RISC-V CPU SoCs.
Expertise in non-coherent and coherent (CHI) NoCs, D2D links, AXI/PCIe fabrics, and GDDR/LPDDR memory systems, backed by performance models correlated with RTL.
Known for driving architecture decisions with clear models, tight cross-functional debug loops, and measurable focus on throughput, latency, utilization, power, and area.
Builds fabric generators, executable specs, and formal checks that turn architecture into verified RTL.

Core Skills

AI Accelerator Architecture: chiplet-based AI/ML accelerator systems, system tradeoff analysis, performance target closure
NoC and Interconnects: non-coherent and coherent (CHI) NoCs, on-package D2D, routing, QoS, topology, clocking, observability, RAS/debuggability
Standards and Fabrics: AXI, CHI, PCIe, cache and fabric integration, heterogeneous SoC connectivity
Memory Systems: GDDR/LPDDR/DDR controllers, memory-channel sizing, interleaving, bandwidth analysis, bottleneck removal
Modeling and Analysis: performance models correlated with RTL, cycle-accurate simulators, workload characterization, latency/bandwidth/utilization studies
Design Automation and Verification: fabric RTL generators, executable specifications, formal verification, AI-assisted RTL
Execution: architecture specifications, design reviews, RTL/PD/verification/software debug, cross-team technical leadership

Experience

Oct 2026 - Present

Tenstorrent, Santa Clara, CA

Staff Architect

Lead the in-house coherent fabric (CHI): transport, chip-to-chip links, and generators.
Fabric lead for a RISC-V CPU chiplet SoC: integrate CPU clusters, the coherent fabric, IOMMU, timers, and an AI compute engine; firmware runs end to end in simulation.
Lead AI-assisted hardware design: executable specs that generate golden models, RTL, and formal checks.
Set performance-modeling direction across the AI NoC and the coherent CPU fabric.
Mentor architects and interns; align modeling, RTL, and verification teams.

May 2025 - Oct 2026

Tenstorrent, Santa Clara, CA

Senior Architect

Chiplet NoC specs: owned non-coherent NoC specs for IO and LPDDR chiplets (topology, address maps, interleaving, KPIs, LPDDR controller width).
NoC RTL and debug: integrated trace, remap, and timeout features in an IO-chiplet fabric; fixed head-of-line blocking to meet bandwidth targets.

GDDR chiplet: drove bring-up, performance analysis, and debug: bandwidth correlation, unaligned-access fixes, and telemetry and timeout checks.
Third-party IP: ran design reviews and evaluations of commercial NoC and memory IP; wrapped vendor models for in-house simulation.
Performance modeling: CPU multicore scaling and SLC policies, AI-accelerator models correlated with RTL, and a cycle-accurate NoC simulator.
Fabric generators: built an AXI fabric generator adopted by SoC teams, and the in-house CHI fabric's generator and formal environment.
Spec platform: built the internal service that builds, hosts, and reviews architecture specs (live editing, comments, AI review); teams use it for chip spec sign-off.
RISC-V ACLINT: generated RTL from an executable spec and checked it formally against the golden model.
Hiring and outreach: built interview frameworks, interviewed candidates, and supported university research collaborations.

May 2023 - August 2023

Tenstorrent, Santa Clara, CA

Platform Architecture Intern

Architected and studied multi-core, multi-chiplet fabric changes to improve performance efficiency and power tradeoffs for server-class compute subsystems.
Estimated on-chip and off-chip traffic for power analysis and early system planning.
Defined cache and fabric verification testplans for heterogeneous integration scenarios.
Built workflows for multi-core and multi-chiplet traffic generation, testing, and simulation.
Configured Arteris IP for non-coherent traffic and exercised corner cases relevant to fabric integration.

May 2022 - August 2022

Tenstorrent, Santa Clara, CA

Platform Architecture Intern

Contributed to architecture studies for a server-class RISC-V multicore CPU.
Developed an in-house configurable fabric performance model for heterogeneous multicore systems.
Defined configuration semantics to represent a broad design space across cores, caches, memory, and fabric options.

Education

2019 - 2025

University of Michigan, Ann Arbor, MI

PhD in Computer Science and Engineering
Thesis: Streamlining High-Performance Heterogeneous Hardware Design

2022 - 2025

University of Michigan, Ann Arbor, MI

Graduate Certificate in Innovation & Entrepreneurship

2016 - 2019

University of Michigan, Ann Arbor, MI

BSc in Computer Science with High Distinction

Selected Publications

2025

Zipper: Latency-Tolerant Optimizations for High-Performance Buses

ASP-DAC 2025
Demonstrated up to 8x speedup for one accelerator with 4.3% logic overhead, and 1.5x speedup for another with 0.9% overhead.

2023

Security Verification of Low-Trust Architectures
CCS 2023, Distinguished Paper Award

2022

Twine: A Chisel Extension for Component-Level Heterogeneous Design
DATE 2022
Introduced a hardware-design methodology for composing heterogeneous systems at the component level.

Service

Program Committee Member, IEEE International Conference on Computer Design (ICCD 2025)
Program Committee Member, Workshop on Computer Architecture Modeling and Simulation (CAMS 2025)

Last updated: October 2026